回顾. 第一章. 非线性波动方程与极化理论.

1. 非线性波动方程.

$$\nabla^2\vec{E}(k,\omega)+k^2\vec{E}(k,\omega)=-\frac{k_0^2}{\varepsilon}\vec{P}_{NL}(k',\omega),$$

$k=k_0 n,\quad k_0=\frac{\omega}{c}$

$$\vec{P}=\varepsilon_0\overleftrightarrow{\chi}^{(1)}\cdot\vec{E}+\underbrace{\varepsilon_0\overleftrightarrow{\chi}^{(2)}\cdot\vec{E}\vec{E}+\varepsilon_0\overleftrightarrow{\chi}^{(3)}\cdot\vec{E}\vec{E}\vec{E}+\cdots}_{\vec{P}_{NL}}$$

2. 非线性极化率. —— 谐振子极化模型.

$$\vec{P}=N\vec{q}(\vec{r}_1+\vec{r}_2+\vec{r}_3+\cdots).$$

3. 非线性极化率 —— 量子力学描述.(微扰理论).

$$i\hbar\frac{\partial}{\partial t}|\psi(t)\rangle=(H_0+H')|\psi(t)\rangle$$

$$|\psi(t)\rangle=\sum_n(a_n^{(0)}+a_n^{(1)}+a_n^{(2)}+\cdots)e^{-i\frac{E_n}{\hbar}t}|n\rangle.$$

4. 非线性极化率的性质:

K-K关系. 对称性. 吸收对极化率的影响.

二阶非线性(三波混频)效应

三波耦合方程.

1. 波动方程 $\longrightarrow$ 各向异性波动方程.

Maxwel eqs: $\nabla \times \vec{E} = -\mu_0 \frac{\partial \vec{H}}{\partial t}$.

$$\nabla \times \vec{H} = \frac{\partial \vec{D}}{\partial t} + \sigma \vec{E}$$

电位移矢量 $\vec{D} = \varepsilon_0 \vec{E} + \vec{P}$.

电极化强度
$$\vec{P} = \varepsilon_0 \overleftrightarrow{\chi}^{(1)} \cdot \vec{E} + \varepsilon_0 \overleftrightarrow{\chi}^{(2)} : \vec{E}\vec{E} + \varepsilon_0 \overleftrightarrow{\chi}^{(3)} : \vec{E}\vec{E}\vec{E} + \cdots$$

$$= \vec{P}^{(1)} + \vec{P}^{(2)} + \vec{P}^{(3)} + \cdots$$

$$= \vec{P}^{(1)} + \vec{P}_{NL}.$$

对非磁性材料，可由上面，得：各项异性非线性时域波动方程.

$$\nabla \times \nabla \times \vec{E}^{NL} + \frac{1}{\varepsilon_0 c^2} \frac{\partial^2}{\partial t^2} \overleftrightarrow{\varepsilon} \cdot \vec{E}^{NL} = -\frac{1}{\varepsilon_0 c^2} \frac{\partial^2 \vec{P}^{NL}}{\partial t^2}$$

均匀、各向同性介质 $\nabla \cdot E = 0$

各向同性、无损耗、$\overleftrightarrow{\varepsilon} \rightarrow \varepsilon_0 (1 + \chi^{(1)}) = \varepsilon_0 n^2$

各向同性、无损耗、非线性时域波动方程：

$$\nabla^2 \vec{E}^{NL} - \frac{n^2}{c^2} \frac{\partial^2 \vec{E}^{NL}}{\partial t^2} = \frac{1}{\varepsilon_0 c^2} \frac{\partial^2 \vec{P}^{NL}}{\partial t^2}$$

说明：

① 散度为0，表现为局部区域（与光波长尺度相当）. 自由与极化电荷为0.

② 无损耗即表现介质对光场不吸收，同时，不吸收也导致介质对光场影响（介电常数）不依赖时间.

③ 各向同性，使介电常数变为标量，但二阶非线性过程一般由非中心对称晶体（各向异性的晶体）产生

考虑，时谐电磁场.

$$\begin{cases} \vec{E}(\vec{r},t) = \sum_n \vec{E}_n(\vec{r}) e^{-i\omega_n t} \\ \vec{P}^{NL}(\vec{r},t) = \sum_n \vec{P}^{NL}_n(\vec{r}) e^{-i\omega_n t} \end{cases}$$

$$\therefore \quad \nabla^2 \vec{E}_n(\vec{r}) + \frac{\omega_n^2}{\varepsilon_0 c^2} \overleftrightarrow{\varepsilon} \cdot \vec{E}_n(\vec{r}) = -\frac{\omega_n^2}{\varepsilon_0 c^2} \vec{P}^{NL}_n(\vec{r})$$

作单色平面波近似.

$$\vec{E}_1(\vec{r},t) = E_1(z) e^{i(k_1 z - \omega_1 t)}$$

$$\vec{E}_2(\vec{r},t) = E_2(z) e^{i(k_2 z - \omega_2 t)}$$

$$P_3 = 4\varepsilon_0 d_{eff} E_1(z) E_2(z) e^{i(k_1+k_2)z}$$

$$\vec{E}_3(\vec{r},t) = E_3(z) e^{i(k_3 z - \omega_3 t)}$$

$$\Delta k = k_1 + k_2 - k_3$$

$$\omega_3 = \omega_1 + \omega_2.$$

有：

$$\frac{d^2 E_3(z)}{dz^2} + 2ik_3 \frac{dE_3(z)}{dz} = -\frac{4 d_{eff} \omega_3^2}{c^2} E_1(z) E_2(z) e^{i\Delta k z}.$$

作 慢变振幅近似：波长尺度下变化很小.

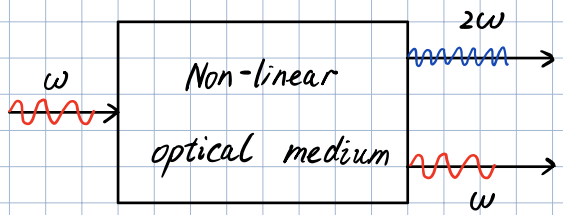
$$\left|\frac{d^2E_3(z)}{dz^2}\right| \longrightarrow \frac{\left|\frac{dE_3(z)}{dz}\right|}{\lambda} \ll \frac{2\pi}{\lambda}\left|\frac{dE_3(z)}{dz}\right|$$

$$\frac{dE_3(z)}{dz} = \frac{2id_{eff}\omega_3^2}{k_3c} E_1(z)E_2(z)e^{i\Delta kz}.$$

考虑三波相互作用，有：三波耦合方程

$$\begin{cases} \dfrac{dE_3(z)}{dz} = \dfrac{2id_{eff}\omega^2}{k_3c^2} E_1(z)E_2(z)e^{i\Delta kz} \\ \dfrac{dE_1(z)}{dz} = \dfrac{2id_{eff}\omega_1^2}{k_1c^2} E_3(z)E_2^*(z)e^{-i\Delta kz} \\ \dfrac{dE_2(z)}{dz} = \dfrac{2id_{eff}\omega_2^2}{k_2c^2} E_3(z)E_1^*(z)e^{-i\Delta kz} \end{cases}$$

小信号近似的二次谐波的产生.



由三波耦合方程:

$$\begin{cases} \dfrac{dE_3(z)}{dz} = \dfrac{2i d_{eff}\omega_3^2}{k_3 c^2} E_1(z) E_2(z) e^{i\Delta k z} \\ \dfrac{dE_1(z)}{dz} = \dfrac{2i d_{eff}\omega_1^2}{k_1 c^2} E_3(z) E_2^*(z) e^{-i\Delta k z} \\ \dfrac{dE_2(z)}{dz} = \dfrac{2i d_{eff}\omega_2^2}{k_2 c^2} E_3(z) E_1^*(z) e^{-i\Delta k z} \end{cases}$$

有: 二次谐波耦合方程.

$$\begin{cases} \dfrac{dE_1(z)}{dz} = \dfrac{2i d_{eff}\omega_1^2}{k_1 c^2} E_2(z) E_1^*(z) e^{-i\Delta k z} \\ \dfrac{dE_2(z)}{dz} = \dfrac{i d_{eff}\omega_2^2}{k_2 c^2} E_1^2(z) e^{i\Delta k z} \end{cases}$$

其中, $\Delta k = 2k_1 - k_2$. $\omega_2 = 2\omega_1$.

作基波小信号近似. $E_1(z) \longrightarrow E_1$.

$$\frac{dE_2(z)}{dz} = \sim \Rightarrow E_2(z) = \sim \int_0^L e^{i\Delta k z} dz.$$

$$\Delta k = 2k_1 - k_2 = 2\frac{2\pi n(\omega_1)}{\lambda_1} - \frac{2\pi n(\omega_2)}{\lambda_2} = 2\frac{\omega_1 n(\omega_1)}{c} - \frac{\omega_2 n(\omega_2)}{c}$$

$$= \frac{\omega_2}{c}[n(\omega_1) - n(\omega_2)]$$

$$\int_0^L e^{i\Delta k z}\,dz = -\frac{i}{\Delta k} e^{i\Delta k z}\Big|_0^L = -\frac{i}{\Delta k} e^{i\Delta kL/2}\left(e^{i\Delta kL/2} - e^{-i\Delta kL/2}\right)$$

$$= L e^{i\Delta kL/2}\,\mathrm{sinc}\,\frac{\Delta kL}{2}.$$

$$E_2(z) = \frac{i d_{eff}\omega_2^2}{k_2 c^2} E_1^2 \cdot \frac{2}{\Delta k} e^{i\Delta kL/2}\sin(\Delta kL/2). \qquad (1)$$

$$I_2 = \frac{1}{2} n_2 \varepsilon_0 c |E_2|^2 = \frac{8 d_{eff}^2 \omega_2^2}{\varepsilon_0 n_1^2 n_2 c^3} I_1^2 \frac{\sin^2\frac{\Delta kL}{2}}{\Delta k^2} \qquad (2)$$

P为功率. S为光斑横截面积.

$$P_2(L) = \frac{8 d_{eff}^2 \omega_2^2}{\varepsilon_0 n_1^2 n_2 c^3}\frac{P_1^2}{S}\frac{\sin^2\frac{\Delta kL}{2}}{\Delta k^2}$$

其中, $\Delta k = \frac{\omega_2}{c}(n_1 - n_2)$. $\omega_i = \frac{2\pi c}{\lambda_i}$ $i = 1, 2$.

转换效率

$$\eta \equiv \frac{P_2}{P_1} = \sim.$$

归一化转化效率.

$$\bar{\eta} \equiv \frac{P_2}{P_1^2} = \sim.$$

$\Delta k = 0$ 情形.

$$\begin{cases} \dfrac{dE_3(z)}{dz} = \dfrac{2id_{eff}\omega_3^2}{k_3c^2} E_1(z)E_2(z)e^{i\Delta kz} \\ \dfrac{dE_1(z)}{dz} = \dfrac{2id_{eff}\omega_1^2}{k_1c^2} E_3(z)E_2^*(z)e^{-i\Delta kz} \\ \dfrac{dE_2(z)}{dz} = \dfrac{2id_{eff}\omega_2^2}{k_2c^2} E_3(z)E_1^*(z)e^{-i\Delta kz} \end{cases}$$

$\Delta k = 2k_1 - k_2 = 0$, $\omega_2 = 2\omega_1$

有2次谐波耦合方程.

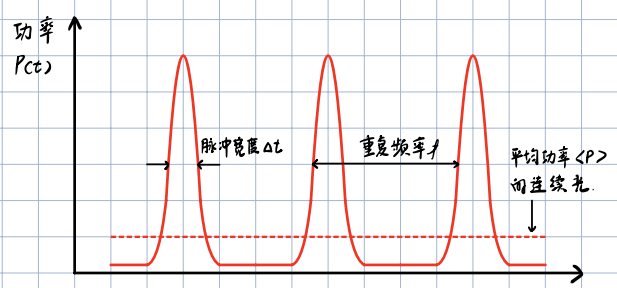
$$\begin{cases} \dfrac{dE_1(z)}{dz} = \dfrac{2id_{eff}\omega_1^2}{kc_1^2} E_2(z)E_1^*(z) \\ \dfrac{dE_2(z)}{dz} = \dfrac{id_{eff}\omega_2^2}{kc_2^2} E_1^2(z) \end{cases}$$

基波小信号近似 $E_1(z) \longrightarrow E_1$

$$\frac{dE_2(z)}{dz} = \frac{id_{eff}\omega_2^2}{k_2c^2} E_1^2$$

$$\Rightarrow \quad E_2(z) = \frac{id_{eff}\omega_2^2}{k_2c^2} E_1^2 L.$$

$$\Rightarrow \quad P_2 = \frac{2d_{eff}^2\omega_2^2}{\varepsilon_0 n_1^2 n_2 c^3} \frac{P_1^2 L^2}{S} \quad \left(I_i = \frac{P_i}{S} = \frac{1}{2} n_i \varepsilon_0 c |E_i|^2 \right).$$



$$P_2(L) = \frac{2d_{eff}^2 \omega_2^2}{\varepsilon_0 n_1^2 n_2 c^3} \frac{P_1^2}{S} \frac{\sin^2\frac{\Delta kL}{2}}{\Delta k^2}$$

看成以 P_{peak} 为功率峰值的矩形波.

$$\langle P\rangle = \int_{\Delta t} P(t)dt \cdot f \quad \Longrightarrow \quad P_{peak} \approx \frac{\langle P\rangle}{\Delta t \cdot f}$$

$$\langle P_2(L)\rangle = \frac{8d_{eff}^2 \omega_2^2}{\varepsilon_0 n_1^2 n_2 c^3} \frac{\langle P_1\rangle^2}{S\Delta t f} \frac{\sin^2\left(\frac{\Delta kL}{2}\right)}{\Delta k^2}$$

转换效率可提高 $\frac{1}{\Delta t f}$ 倍.

3.3 二阶非线性极化系数及其张量.

二阶非线性系数 $\chi^{(2)}$ 基本性质

1. 二阶非线性极化系数.

简谐振子模型

$$\frac{d^2\vec{x}}{dt^2} + 2\gamma\frac{d\vec{x}}{dt} + \omega_0^2\vec{x} + a\vec{x}^2 = \frac{q\vec{E}(t)}{m}.$$

$$\vec{F}_{restoring} = -m\omega_0^2\vec{x} - ma\vec{x}_0^2$$ 非简谐力.

作微扰近似:

$$\vec{P} = \varepsilon_0\chi^{(1)}\vec{E} + \varepsilon_0\chi^{(2)}\vec{E}\vec{E} + \varepsilon_0\chi^{(3)}\vec{E}\vec{E}\vec{E} + \cdots$$

$$= Nq(\vec{x}^{(1)} + \vec{x}^{(2)} + \vec{x}^{(3)} + \cdots).$$

单位体积电荷数.

二阶非线性效应

$$\tilde{E}(t) = E_1 e^{-i\omega_1 t} + E_2 e^{-i\omega_2 t} + c.c$$

$$\chi^{(1)}(\omega_j) = \frac{Nq^2}{\varepsilon_0 m D(\omega_j)} \quad (j=1,2).$$

$$D(\omega) = \omega_0^2 - \omega^2 - 2i\omega\gamma.$$

$$\chi^{(2)}(2\omega_1;\omega_1,\omega_1) = \frac{aNq^3}{\varepsilon_0 m^2}\frac{1}{D(2\omega_1)D(\omega_1)D(\omega_1)}$$

$$\chi^{(2)}(\omega_1+\omega_2;\omega_1,\omega_2) = \frac{aNq^3}{\varepsilon_0 m^2}\frac{1}{D(\omega_1+\omega_2)D(\omega_1)D(\omega_2)}$$

$$\chi^{(2)}(\omega_1-\omega_2;\omega_1,-\omega_2)=\frac{aNq^3}{\varepsilon_0 m^2}\frac{1}{D(\omega_1-\omega_2)D(\omega_1)D(-\omega_2)}$$

$$\chi^{(2)}(0;\omega_1,-\omega_1)=\frac{aNq^3}{\varepsilon_0 m^2}\frac{1}{D(0)D(\omega_1)D(-\omega_1)}$$

线性极化率：

$$\chi^{(1)}(\omega)=\frac{Nq^2}{\varepsilon_0 m D(\omega)}=Re(\chi^{(1)}(\omega))+i\cdot Im(\chi^{(1)}(\omega)).$$

$$D(\omega)=\omega_0^2-\omega^2-2i\omega\gamma.$$

共振频率　驱动频率.

$$\begin{cases} Re(\chi^{(1)}(\omega))=\dfrac{Nq^2}{\varepsilon_0 m}\dfrac{\omega_0^2-\omega^2}{(\omega_0^2-\omega^2)^2+4\omega^2\gamma^2} \\ Im(\chi^{(1)}(\omega))=\dfrac{Nq^2}{\varepsilon_0 m}\dfrac{2\gamma\omega}{(\omega_0^2-\omega^2)^2+4\omega^2\gamma^2} \end{cases}$$

- 损耗≠色散，在远离共振区的两端线性极化率不同，低频无色散.
- $\vec{\varepsilon}=\varepsilon_0[1+\vec{\chi}^{(1)}]$. 如何修改以上公式以体现$\chi^{(1)}$的各向异性?.

二阶非线性系数对称性.

处于共振波长处增强二阶非线性系数但同时引入吸收.

$$\chi^{(2)}(\omega_3;\omega_1,\omega_2)=\frac{aNg^2}{\varepsilon_0 m^2}\frac{1}{D(3)\,D(2)\,D(1)}$$

$$D(\omega)=\omega_0^2-\omega^2-2i\omega\gamma$$

$$=\frac{a\varepsilon_0^2 m^3}{N^2 g^3}\chi^{(1)}(\omega_3)\chi^{(1)}(\omega_1)\chi^{(1)}(\omega_2).$$

$$\omega_3=\omega_1+\omega_2$$

本征置换对称性：

- 入射场相互交换，即后2个下标与频率同时交换.
- 共振与非共振区均成立.

$$\chi^{(2)}_{ijk}(\omega_3;\omega_1,\omega_2)=\chi^{(2)}_{ijk}(\omega_3;\omega_2,\omega_1)=\frac{aNg^3}{\varepsilon_0 m^2}\frac{1}{D(\omega_3)D(\omega_1)D(\omega_2)}$$

全互换对称性

- 所有电场全部交换，全部下标与频率全部交换：
- 非共振区成立，无损耗.

$\omega_0 \gg \omega_1,\omega_2,\omega_3$. or $\omega_0 \ll \omega_1,\omega_2,\omega_3$.

$$\chi^{(2)}_{ijk}(\omega_3;\omega_1,\omega_2)=\chi^{(2)}_{ijk}(\omega_1;\omega_3,-\omega_2)=\chi^{(2)}_{ijk}(\omega_2;\omega_3,-\omega_1)=\cdots$$

$$=\frac{aNg^3}{\varepsilon_0 m^2}\frac{1}{(\omega_0^2-\omega_1^2)(\omega_0^2-\omega_2^2)(\omega_0^2-\omega_3^2)}$$

复共轭对称性.

- 共振与非共振区均成立.

$$[\chi^{(2)}_{ijk}(\omega_3;\omega_1,\omega_2)]^* = [\chi^{(2)}_{ijk}(-\omega_3;-\omega_1,-\omega_2)].$$

时间反演对称性.

- 非共振区成立, 无损耗.

$$\chi^{(2)}(\omega_3;\omega_1,\omega_2) = [\chi^{(2)}(-\omega_3;-\omega_1,-\omega_2)].$$

二阶非线性极化矩阵的运算.

$$\begin{pmatrix} P_1(2\omega) \\ P_2(2\omega) \\ P_3(2\omega) \end{pmatrix} = \varepsilon_0 \begin{pmatrix} 0 & 0 & 0 & 0 & \chi_{15} & -\chi_{22} \\ -\chi_{22} & \chi_{22} & 0 & \chi_{15} & 0 & 0 \\ \chi_{31} & \chi_{31} & \chi_{33} & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} E_1^2(\omega) \\ E_2^2(\omega) \\ E_3^2(\omega) \\ 2E_2(\omega)E_3(\omega) \\ 2E_1(\omega)E_3(\omega) \\ 2E_1(\omega)E_2(\omega) \end{pmatrix}.$$